

Linearni tipi

Vida Mlinar

Naravna dedukcija

Naravna dedukcija

$$\frac{A \quad B}{A \wedge B} \wedge\text{-intro}$$

Naravna dedukcija

$$\frac{\frac{A}{\frac{A \wedge B}{A} \wedge_1\text{-elim}} \quad \frac{B}{\frac{A \wedge B}{B} \wedge_2\text{-elim}}}{A \wedge B} \wedge\text{-intro}$$

Naravna dedukcija

$$\frac{\frac{A}{\frac{A \wedge B}{A} \wedge_1\text{-elim}} \quad \frac{B}{\frac{A \wedge B}{B} \wedge_2\text{-elim}}}{A \wedge B} \wedge\text{-intro}$$

Naravna dedukcija

$$\frac{\frac{A}{A \wedge B} \wedge_1\text{-elim} \quad \frac{B}{A \wedge B} \wedge_2\text{-elim}}{A \wedge B} \wedge\text{-intro}$$

$$\frac{A \Rightarrow B}{A} \Rightarrow\text{-elim}$$

Naravna dedukcija

$$\frac{\frac{A}{A \wedge B} \wedge_1\text{-elim} \quad \frac{B}{A \wedge B} \wedge_2\text{-elim}}{A \wedge B} \wedge\text{-intro}$$

$$\frac{A \vdash B}{A \Rightarrow B} \Rightarrow\text{-intro} \quad \frac{A \Rightarrow B \quad A}{B} \Rightarrow\text{-elim}$$

Naravna dedukcija

$$\frac{\frac{\Gamma \vdash A \quad \Gamma \vdash B}{\Gamma \vdash A \wedge B} \wedge\text{-intro}}{\frac{\Gamma \vdash A \wedge B}{\Gamma \vdash A} \wedge_1\text{-elim} \quad \frac{\Gamma \vdash A \wedge B}{\Gamma \vdash B} \wedge_2\text{-elim}}$$

$$\frac{\Gamma, A \vdash B}{\Gamma \vdash A \Rightarrow B} \Rightarrow\text{-intro} \quad \frac{\Gamma \vdash A \Rightarrow B \quad \Gamma \vdash A}{\Gamma \vdash B} \Rightarrow\text{-elim}$$

Naravna dedukcija

$$\frac{\frac{\Gamma \vdash A \quad \Gamma \vdash B}{\Gamma \vdash A \wedge B} \wedge\text{-intro}}{\frac{\Gamma \vdash A \wedge B}{\Gamma \vdash A} \wedge_1\text{-elim} \quad \frac{\Gamma \vdash A \wedge B}{\Gamma \vdash B} \wedge_2\text{-elim}}$$

$$\frac{\Gamma, A \vdash B}{\Gamma \vdash A \Rightarrow B} \Rightarrow\text{-intro} \quad \frac{\Gamma \vdash A \Rightarrow B \quad \Gamma \vdash A}{\Gamma \vdash B} \Rightarrow\text{-elim}$$

$$\frac{A \in \Gamma}{\Gamma \vdash A} \text{hip}$$

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$$\frac{}{\Gamma, A \vdash A} \text{hip}$$

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$$\frac{}{A \vdash A} \text{hip}$$

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$$\frac{}{A \vdash A} \text{hip}$$

$$\frac{\Gamma, A \vdash B}{\Gamma \vdash A \multimap B} \multimap\text{-intro}$$

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$$\frac{}{A \vdash A} \text{hip}$$

$$\frac{\Gamma, A \vdash B}{\Gamma \vdash A \multimap B} \multimap\text{-intro}$$

$$\frac{\Gamma \vdash A \multimap B \quad \Delta \vdash A}{\Gamma, \Delta \vdash B} \multimap\text{-elim}$$

Linearna logika

$$\frac{}{A \vdash A} \text{hip}$$

$$\frac{\Gamma, A \vdash B}{\Gamma \vdash A \multimap B} \multimap\text{-intro}$$

$$\frac{\Gamma \vdash A \multimap B \quad \Delta \vdash A}{\Gamma, \Delta \vdash B} \multimap\text{-elim}$$

Curry-Howardov izomorfizem

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$$\frac{x : A \in \Gamma}{\Gamma \vdash x : A} \text{ var}$$

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$$\frac{x : A \in \Gamma}{\Gamma \vdash x : A} \text{ var}$$

$$\frac{\Gamma, x : A \vdash M : B}{\Gamma \vdash (x \mapsto M) : A \rightarrow B} \rightarrow\text{-intro}$$

Curry-Howardov izomorfizem

$$\frac{x : A \in \Gamma}{\Gamma \vdash x : A} \text{ var}$$

$$\frac{\Gamma, x : A \vdash M : B}{\Gamma \vdash (x \mapsto M) : A \rightarrow B} \rightarrow\text{-intro}$$

$$\frac{\Gamma \vdash M : A \rightarrow B \quad \Gamma \vdash N : A}{\Gamma \vdash MN : B} \rightarrow\text{-elim}$$

Curry-Howardov izomorfizem

$$\frac{x : A \in \Gamma}{\Gamma \vdash x : A} \text{ var}$$

$$\frac{\Gamma, x : A \vdash M : B}{\Gamma \vdash (x \mapsto M) : A \rightarrow B} \rightarrow\text{-intro}$$

$$\frac{\Gamma \vdash M : A \rightarrow B \quad \Gamma \vdash N : A}{\Gamma \vdash MN : B} \rightarrow\text{-elim}$$

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$$\frac{}{x : A \vdash x : A} \text{var}$$

$$\frac{\Gamma, x : A \vdash M : B}{\Gamma \vdash (x \mapsto M) : A \multimap B} \multimap\text{-intro}$$

$$\frac{\Gamma \vdash M : A \multimap B \quad \Delta \vdash N : A}{\Gamma, \Delta \vdash MN : B} \multimap\text{-elim}$$

Linearni tipi

$$\frac{}{x : A \vdash x : A} \text{var}$$

$$\frac{\Gamma, x : A \vdash M : B}{\Gamma \vdash (x \mapsto M) : A \multimap B} \multimap\text{-intro}$$

$$\frac{\Gamma \vdash M : A \multimap B \quad \Delta \vdash N : A}{\Gamma, \Delta \vdash MN : B} \multimap\text{-elim}$$